

Dear FCUSD Team, Dr. Harsh, and everyone involved,

Following today's discussion with the FCUSD student team, I wanted to summarize our current understanding of the AI-driven emergency response project so that everyone has a clear picture of the proposed direction, ongoing activities, and the roles that different groups may take up moving forward.

Project Vision

The overall objective is to combine **PeopleSense occupancy intelligence, emergency alerting**, and **AI-driven decision support** into a unified system that can assist emergency managers during developing incidents.

At a high level, the system would receive an emergency trigger (for example, a CAP-OIP alert generated through the PeopleSense rules engine or real-time WIS CAP feeds), having enriched occupancy-related information from PeopleSense-monitored properties, and then leverage a collection of AI agents to generate actionable insights regarding evacuation and response.

Current Team Structure

At present, we have identified two major workstreams:

1. Data Modelling and API Team

- Data collection and exploration from [Kaggle](#) and other publicly available evacuation-related datasets.
- Development of evacuation-related data models.
- Creation of APIs and Python services that expose model functionality.
- Implementation of supporting data-processing logic.

2. AI Agent Team

- Design of agent orchestration mechanisms.
- Development of specialized AI agents and their interactions.
- Integration of domain-specific Python functionality with agents.
- Exploration of decision-support workflows and agent collaboration strategies.

Proposed Data Modelling Approach

The evacuation datasets that we have explored so far contain several useful variables related to evacuation outcomes. The primary attributes we intend to utilize include:

- **Scenario Type**

- Earthquake
- Flood
- Fire
- Tsunami
- Tornado
- Other emergency scenarios
- **Occupancy Count**
 - Number of individuals present during the event.
- **Evacuation Time**
 - Time required to evacuate the occupants successfully.
- **Category**
 - Ideally, we would have preferred categories describing property types (e.g., office building, stadium, park, transit facility, etc.).
 - However, the currently available datasets do not consistently provide such information.
 - Instead, they include variables such as:
 - Sensor type,
 - Evacuation category,
 - Success percentage, etc.
 - Therefore, for the initial phase, we may select one of these existing dimensions as a proxy category. For example, success percentage may be grouped into:
 - Below 95%,
 - Between 95% and 97.5%,
 - Above 97.5%.

The purpose of developing this model is to estimate expected evacuation durations under varying emergency scenarios and occupancy conditions.

Proposed Operational Flow

When an emergency event occurs:

1. PeopleSense detects or receives a triggering event.
2. A CAP alert is generated/fetched.
3. Occupancy-related information associated with affected PeopleSense properties is attached to the alert. (We may switch to other occupancy data as well)
4. The alert is forwarded to an AI orchestration layer.
5. Multiple AI agents perform specialized tasks in parallel.
6. One such agent, tentatively referred to as the **Evacuation Expert Agent**, estimates evacuation requirements.

The Evacuation Expert Agent would:

- Identify the relevant emergency scenario,
- Determine the applicable category,
- Retrieve occupancy information,
- Invoke a Python service through an API,
- Obtain estimated evacuation durations from the evacuation model,
- Refine those estimates based on contextual considerations,
- Share the resulting information with other agents for further reasoning.

The specific responsibilities of additional agents remain an open discussion topic for upcoming meetings.

Potential Outputs

Some of the potential outputs of this system include:

- Decision-support recommendations for emergency managers,
- Suggested evacuation strategies,
- Operational guidance during active incidents,
- Automated summaries describing how events unfolded,
- Incident reports documenting:
 - the time of occurrence,
 - occupancy conditions,
 - recommended actions,
 - evacuation estimates,
 - strategies employed,
 - eventual outcomes.

Such outputs could later serve both operational purposes and retrospective analysis as our data models and AI capabilities mature.

Current Assignments

Based on today's discussion:

- **Praneeth** and **Ojas** have decided to contribute primarily toward the **AI modelling and agent development workstream**.
- **Alex** has started work on **data exploration and evacuation data modelling**.

As additional students join the effort, they may align themselves with either team depending on their interests and expertise so that both workstreams progress in parallel.

Next Steps

Tomorrow, we plan to reconvene and discuss individual progress updates from both groups. The goal will be to consolidate our understanding before presenting the approach to **Larry on Thursday**, who can help guide the broader direction of this initiative.

We would also greatly appreciate feedback from **Larry** and **Dr. Verma** regarding:

- the proposed architecture,
- assumptions being made within the data models,
- agent responsibilities,
- opportunities for refinement or simplification.

In the meantime, I will continue coordinating across both student groups to help manage activities, facilitate communication, and support progress on both the data and AI components.

Please feel free to reply on this email thread if you have any questions, suggestions, concerns, or ideas that you believe should be incorporated into the project direction.

Thank you everyone for your time and contributions. I look forward to our discussion tomorrow. Attached is a simple mermaid/flow diagram created for the project above (Nothing is final right now and this is only created for understanding purposes).

